Universalities in BCS Theory

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Joint with Joscha Henheik and Barbara Roos

Garmisch
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Bardeen–Cooper–Schrieffer (BCS) 1957 theory of superconductivity

Described by **BCS gap equation** in $d = 1, 2, 3$ dimensions

$$\Delta(p) = -\frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} \hat{V}(p-q) \frac{\Delta(q)}{K_T^{\Delta}(q)} \, \mathrm{d}q,$$

with

$$K_T^{\Delta}(p) = \frac{E_\Delta(p)}{\tanh\left(\frac{E_\Delta(p)}{2T}\right)}, \qquad E_\Delta(p) = \sqrt{(p^2-\mu)^2 + |\Delta(p)|^2}.$$

Electron-lattice interaction leads to effective attractive electron-electron interaction V.

Material is superconducting if non-zero solution Δ exists.

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Arises as Euler–Lagrange equation of **BCS functional**

$$\mathcal{F}(\gamma, \alpha) = \int_{\mathbb{R}^d} (p^2 - \mu)\gamma(p) \, \mathrm{d}p + \int_{\mathbb{R}^d} V(x) |\alpha(x)|^2 \, \mathrm{d}x - TS(\gamma, \alpha).$$

Grand canonical pressure functional restricted to translation- and $SU(2)$-invariant (i.e. spin singlet) quasi-free states and neglecting "direct" and "exchange" terms.

- γ is particle density at momentum p, $\gamma(p) \sim \left\langle a_{p\uparrow}^\dagger a_{p\uparrow} \right\rangle$
- α is Cooper-pair wave function, $\hat{\alpha}(p) \sim \left\langle a_{p\uparrow}^\dagger a_{-p\downarrow}^\dagger \right\rangle$

Theorem (Hainzl–Hamza–Seiringer–Solovej 2008)

Let $V \in L^{p_d}$ be real-valued and reflection-symmetric ($p_3 = 3/2$, $p_2 > 1$, $p_1 = 1$), $\mu \in \mathbb{R}$ and $0 \leq T < \infty$. Then the following are equivalent

1. *A minimizer of the BCS functional has $\alpha \neq 0$,*
2. *There exists $\alpha \neq 0$ with $(K_T^{\Delta} + V)\alpha = 0$,*
   (equivalently $\Delta = -2\widehat{V\alpha}$ satisfies the BCS gap equation),
3. *The operator $K_T^0 + V$ has at least one negative eigenvalue.*

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$\rightsquigarrow$ Define the **critical temperature** $T_c := \inf\{T \geq 0 : K_T^0 + V \geq 0\}$.

Phase transition at T_c: superconducting for $T < T_c$ and normal metal for $T \geq T_c$. (Follows easily by K_T monotone increasing in T.)

$E_\Delta(p) = \sqrt{(p^2 - \mu)^2 + |\Delta(p)|^2}$ dispersion relation for associated quasi-particles.

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T_c and Ξ are material-dependent (μ, V-dependent). Both measure "stability" of superconducting phase.

Question

- Formula for T_c and Ξ in various limits?
- Relation between T_c and Ξ ?

Zero temperature

Physics prediction for T_c small:

$$\frac{\Xi(T=0)}{T_c} \approx \pi e^{-\gamma} \approx 1.76$$

Well-known in physics literature.

Valid for s-wave superconductivity (Δ radial),
(Ex: V radial with $\hat{V} \leq 0$)

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- Weak coupling λV with $\lambda \to 0$,
- Low density $\mu \to 0$,
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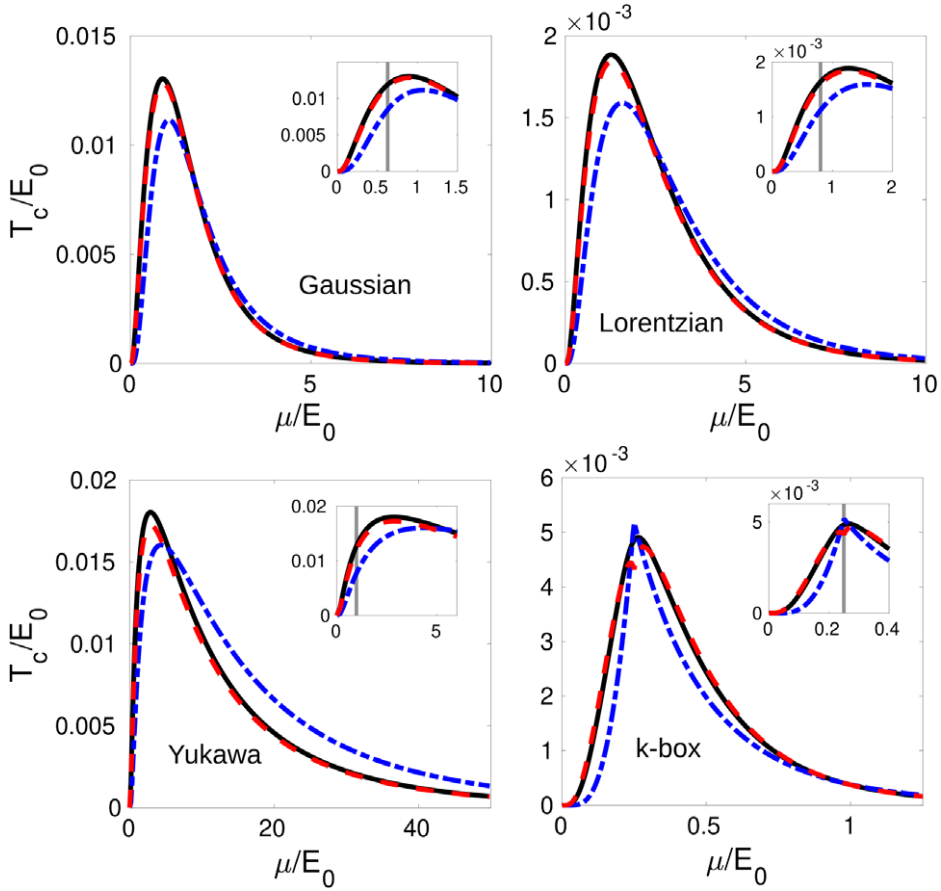

Figure: From "E. Langmann, C. Triola, and A. V. Balatsky. Phys. Rev. Lett. **122**.15 (2019)"

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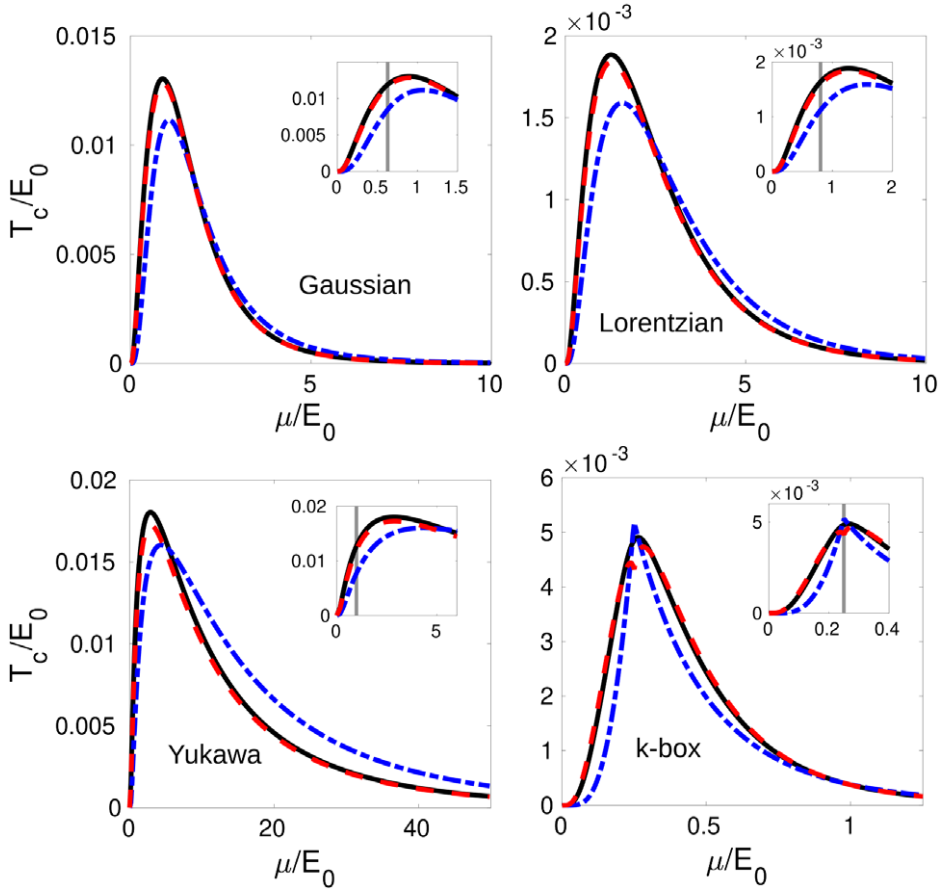
Can we find T_c and $\Xi(T=0)$ in these limits?



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In all limits: $$\frac{\Xi(T=0)}{T_c} = \pi e^{-\gamma}(1 + o(1))$$

"General" result for $d = 3$ and s-wave superconductivity [Langmann–Triola–Balatsky 2019; Langmann–Triola 2023].

Non-zero temperature

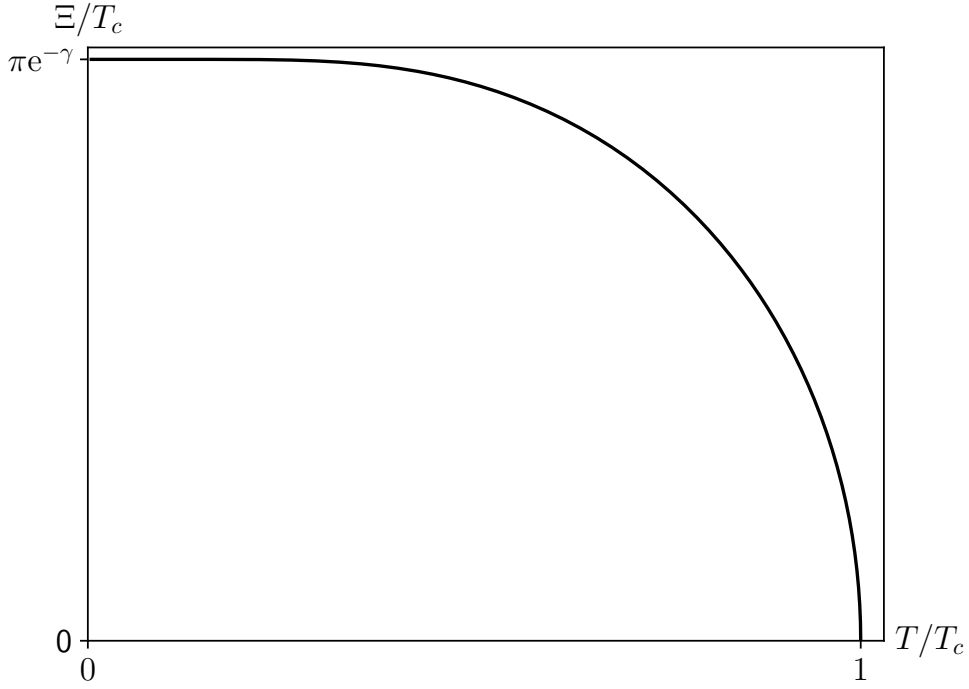
Physics prediction for T_c small (for s-wave superconductivity):

$$\frac{\Xi(T)}{T_c} \approx \mathrm{f}_{\mathrm{BCS}}\left(\sqrt{1-\frac{T}{T_c}}\right)$$

for **universal** function $\mathrm{f}_{\mathrm{BCS}}$.
For $T=0$ and $T \to T_c$:

$$\frac{\Xi(T=0)}{T_c} \approx \pi e^{-\gamma} \approx 1.76,$$

$$\frac{\Xi(T)}{T_c} \approx \sqrt{\frac{8\pi^2}{7\zeta(3)}}\sqrt{1-\frac{T}{T_c}} \approx 3.06\sqrt{1-\frac{T}{T_c}}$$



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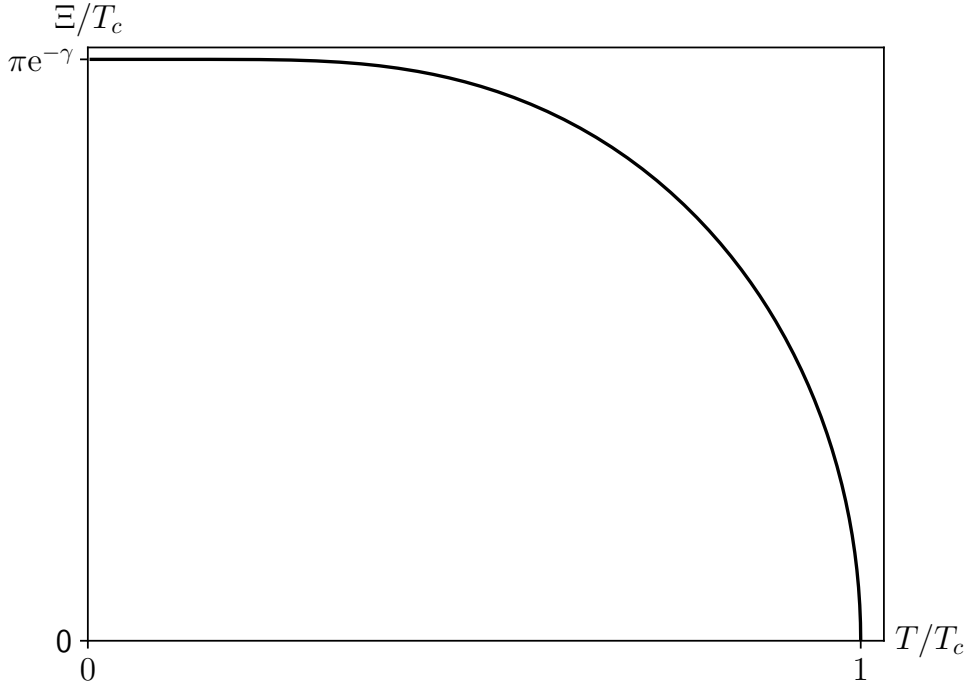

Theorem (Langmann–Triola 2023, Henheik–L. 2023)

Picture holds

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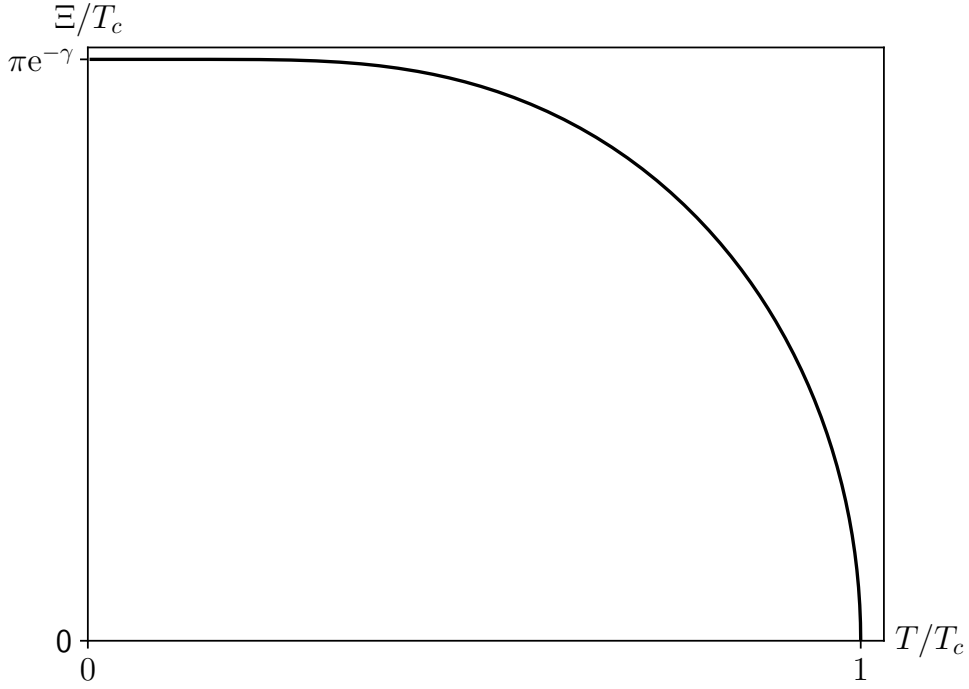

Theorem (Langmann–Triola 2023, Henheik–L. 2023)

Picture holds for interaction λV radial in limit $\lambda \to 0$

- *in dimensions $d = 1, 2, 3$ for V with $\hat{V} \leq 0$*
  (s-wave superconductivity, weak coupling limit),
- *in dimension $d = 2$ for V with* $\dim\ker(K_{T_c} + \lambda V) \leq 2$
  (pure angular momentum, weak coupling limit),

Main ideas of proofs — Birman–Schwinger principle

"$\Delta \neq 0$ solves BCS gap equation"

$\Leftrightarrow$ "$K_T^\Delta + V$ has lowest eigenvalue 0 and associated eigenvector $\alpha = -\frac{\Delta}{2K_T^\Delta}$"

$\Leftrightarrow$ "Birman–Schwinger operator

$$B_{T,\Delta} = \operatorname{sgn} V |V|^{1/2} (K_T^\Delta)^{-1} |V|^{1/2}$$

has -1 as lowest eigenvalue and associated eigenvector $\phi = \operatorname{sgn} V |V|^{1/2} \alpha$."

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To study $B_{T,\Delta}$ define

$$m(T,\Delta) = \frac{1}{|\mathbb{S}^{d-1}|} \int_{|p| \leq \sqrt{2\mu}} \frac{1}{K_T^\Delta(p)} \, \mathrm{d}p$$

Decompose (with $\mathfrak{F}_\mu : L^1(\mathbb{R}^d) \to L^2(\mathbb{S}^{d-1})$ a rescaled Fourier transform)

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$$m(T=0,\Delta) \approx \mu^{d/2-1}\left[\log\frac{\mu}{\Xi(T=0)} + \log(2c_d)\right],$$

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Positive temperature: Neglect error term $M_{T,\Delta}$ and consider for T and $T=T_c$:

$$\int_0^{\sqrt{2\mu}}\left(\frac{1}{K_T^\Delta(p)} - \frac{1}{K_{T_c}^0(p)}\right)|p|^{d-1}\,\mathrm{d}p = m(T,\Delta) - m(T_c,\Delta=0) \approx 0$$

Essentially defining equation for $\mathrm{f}_{\mathrm{BCS}} \rightsquigarrow \Xi(T)/T_c \approx \mathrm{f}_{\mathrm{BCS}}(\sqrt{1-T/T_c})$.